

Kobe Barrette

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Projects

Portable Air Quality Monitor Device

Tools: Jupyter Notebook, Python, GitHub, Kotlin, C

- Built a wearable device to monitor a person's immediate air quality surroundings.
- Developed a machine learning model for Particulate Matter classification.
- Developed an Android application with Bluetooth device communication.

Glances Open-Source Project

Tools: Python, VS code, GitHub

- Developed new-feature requests for Glances open-source monitoring application.
- Reverse engineered the project using UML diagrams.

Project Tracking Dashboard

Tools: Java, Python, SQL, CSS, Ignition

- Designed a website using Ignition SCADA's perspective module to track company projects in real-time.
- Developed and integrated database model.

Experience

J.L. Richards & Associates Limited- Electrical Student (Co-op)

May 2025-Aug 2025

- Company Overview: A large engineering consulting firm, comprising 7 offices around Ontario, offering services in architectural and engineering disciplines
- Worked as part of the electrical team to design and develop systems for industrial and commercial applications, leveraging AutoCAD, and electrical theory

Northern Ontario Wires Inc - Engineering Research and Development (Co-op)

May 2024-Aug 2024

- Company Overview: A local electricity distribution company responsible for distributing power the towns of Cochrane, Iroquois Falls and Kapuskasing.
- Developed an Excel tool to assist in creating parts lists for new lines being constructed. The tool uses Visual Basics and follows Utility Standards Forum (USF).

RidgeTech Automation - Control Systems Specialist (Co-op)

Jan 2024-Apr 2024

- Company Overview: A control system developer and consulting company having provided solutions for over \$1 billion worth of industrial equipment.
- Worked as part of the integrated software team to develop high quality applications for various clients using Ignition SCADA.
- Adapted an Agile Scrum framework participating in daily stand-ups, weekly demo sessions, and managing jobs using Atlassian's JIRA.
- Performed networking tasks such as server configuration using Proxmox Virtual Environment.

Department of National Defence - Engineering Student Project Manager (Co-op)

May 2023-Aug 2023

- Worked on the Chemical Biological Radiological Nuclear explosives Enhancement (CBRNe) project to procure advanced (multi-million dollar) engineering capabilities for the Canadian Special Operations Forces Command (CANSOFCOM).
- Researched complex mechanical systems, some of which include night vision goggles, naval remote turret systems and autonomous explosive drones.

Education

University of Guelph

2021-Present

- Bachelor of Engineering, Engineering Systems and Computing
- GPA: 84%